

			Marks	CO	PO	Level
Q.1	1.	Explain the von Neumann architecture and describe how it influences the design of programming languages.	5	1	1,2,3,4	2
	2.	Orthogonality in programming languages is often considered a desirable property. Explain what orthogonality means in this context. Then, analyze a scenario where high orthogonality could lead to both benefits and challenges in language design. Support your answer with examples.	4	2	1,2,3,4,5	4
Q.2	1.	Explain phases of compilers in detail.	4	1	1,2,3,4	3
	2.	float rate = (base + bonus) / 2.0e-3; // calculate adjusted rate Perform lexical analysis on the above statement in detail.	4	2	1,2,3,4,5	3
	3	Differentiate between a parse tree and an abstract syntax tree (AST). How would the AST for $3 + 4 * 5$ differ from the parse tree?"	3	2	1,2,3,4,5	3
Q.3	1.	Discuss code optimisation process. Demonstrate code optimisation with the help of example.	4	3	1,2,3,4,5,8	4



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2	Explain the purpose of Backus–Naur Form (BNF) in defining the syntax of programming languages. Then, using BNF, define the grammar for a simple arithmetic expression that includes addition and multiplication of integer literals. Show at least two rules with proper notation.	4	3	1,2,3,4,5,8	4
3	Define semantics in the context of programming languages and explain how it differs from syntax. Illustrate your explanation by describing what kind of errors can occur if semantic rules are violated, giving one example.	2	3	1,2,3,4,5,8	4

*****Best of Luck*****